

Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0
Group Marks :	706
Is this Group for Examiner? :	No
Examiner permission :	Cant View
Show Progress Bar? :	No
Revisit allowed for group Instructions? :	Yes
Maximum Instruction Time :	0
Minimum Instruction Time :	0
Group Time In :	Minutes
Revisit Section :	Yes
Action on Revisit Section :	View and Edit
Navigate To Group Summary From Last Question? :	No
Disable Submit Button During Assessment? :	No
Section Selection Time? :	0
No of Optional sections to be attempted :	0

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Section Id :	64065374330
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653157501
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406531031597 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SOFTWARE ENGINEERING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406533488818. ✓ YES

6406533488819. ✗ NO

Sub-Section Number :

2

Sub-Section Id :

640653157502

Question Shuffling Allowed :

Yes

Question Number : 2 Question Id : 6406531031602 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A software organization planned to develop a mobile application that can be used by any other organization for its employees to manage their leaves and holidays efficiently. The software organization interviewed employees from different organizations to collect requirements. In this case, which type of client is the software organization dealing with?

Options :

6406533488833. ✓ external user

6406533488834. ✗ internal user

6406533488835. ✗ a software or a software component

6406533488836. ✗ none of these

Question Number : 3 Question Id : 6406531031603 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A software company has implemented a website for a school. The website is given to the school faculty members to verify if the website meets the requirements provided by the school. The faculty members use the website and find that when they try to upload results of more than 1000 students, the website crashes. Which phase of the SDLC is the software currently in?

Options :

6406533488837. ✗ Requirement gathering

6406533488838. ✗ Development phase

6406533488839. ✗ Design phase

6406533488840. ✓ Testing phase

6406533488841. ✖ Maintenance phase

Question Number : 4 Question Id : 6406531031604 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following two requirements for a school portal:

Requirement 1: The portal should allow bulk upload of the students' marks.

Requirement 2: The portal should allow a student to download his or her marksheet within 30 seconds. Which of the following statements is **true**?

Options :

6406533488842. ✔ Requirement 1 is a functional requirement, whereas Requirement 2 is a non functional requirement.

6406533488843. ✖ Requirement 2 is a functional requirement, whereas Requirement 1 is a non functional requirement.

6406533488844. ✖ Both requirements are functional requirements.

6406533488845. ✖ Both requirements are non functional requirements.

Question Number : 5 Question Id : 6406531031605 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Which of the following UML diagrams models flow of control among computational activities?

Options :

6406533488846. ✖ Class diagram

6406533488847. ✖ State machine view

6406533488848. ✔ Activity view

6406533488849. ✖ Interaction view

Question Number : 6 Question Id : 6406531031606 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Which of the following software can be used for sharing and maintaining code versions?

Options :

6406533488850. ✖ Vim

6406533488851. ✔ Git

6406533488852. ✖ Flask

6406533488853. ✖ Visual Studio Code

Question Number : 7 Question Id : 6406531031607 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A prototype is presented as a web application along with some fabricated data.

Determine the prototype type that has been provided.

Options :

6406533488854. ✖ Storyboards

6406533488855. ✖ Paper prototype

6406533488856. ✖ Digital mock-ups

6406533488857. ✔ Interactive prototype

Sub-Section Number :

3

Sub-Section Id :

640653157503

Question Shuffling Allowed :

Yes

Question Number : 8 Question Id : 6406531031608 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Suppose the risk assessed for personnel shortfall in a given software project is 1.8. The probability that personnel shortage will occur during project development is 0.6. What is the impact of personnel shortage on the given project?

Options :

6406533488858. ✖ 1.08

6406533488859. ✖ 2

6406533488860. ✔ 3

6406533488861. ✖ 0.33

Question Number : 9 Question Id : 6406531031609 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider a module *Deque* that maintains a doubly linked list. The module *Deque* has the following functions:

- push_back(e) : adds an element *e* at the back of the doubly linked list,

- push_front(e) : adds an element *e* at the front of the doubly linked list,

- pop_back(): removes an element from back of the doubly linked list,

- pop_front(): removes an element from front of the doubly linked list.

Identify the type of cohesion for the module *Deque*.

Options :

6406533488862. ✖ Sequential cohesion

6406533488863. ✔ Communicational cohesion

6406533488864. ✖ Procedural cohesion
6406533488865. ✖ Coincidental cohesion

Question Number : 10 Question Id : 6406531031610 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Match the following regarding the outcomes of software design process.

Outcomes	Descriptions
1. Components	A. Designed for storing and managing data
2. Interfaces	B. Designed to accomplish some well-defined tasks
3. Data structures	C. Required to implement individual components
4. Algorithms	D. Define how components communicate with each other

Options :

6406533488866. ✖ 1-A, 2-B, 3-D, 4-C
6406533488867. ✖ 1-C, 2-B, 3-D, 4-A
6406533488868. ✖ 1-B, 2-A, 3-D, 4-C
6406533488869. ✔ 1-B, 2-D, 3-A, 4-C

Question Number : 11 Question Id : 6406531031611 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider an eCommerce portal that has integrated a set of APIs to recommend products to customers based on their carts. Which of the following debugging strategies involves the task of searching the documentation and code examples to understand the correct use of the APIs?

Options :

6406533488870. ✖ Input manipulation
6406533488871. ✖ Backwards
6406533488872. ✖ Forwards
6406533488873. ✔ Blackbox debugging

Question Number : 12 Question Id : 6406531031612 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider the Python code below.

```
def binsearch(arr, key):
    l = 0
    h = len(arr) - 1
    while l <= h:
        m = (l + h) // 2
        if arr[m] == key:
            return m
        else:
            if arr[m] < key:
                l = m + 1
            else:
                h = m - 1
    return -1
```

What is the cyclomatic complexity of this code?

Options :

6406533488874. ✖ 2

6406533488875. ✖ 3

6406533488876. ✔ 4

6406533488877. ✖ 5

Question Number : 13 Question Id : 6406531031613 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider designing an online gift store that delivers various gift items in a variety of ways, such as direct pick-up from the store, delivery by its own logistics team, or delivery by other delivery partners. To support the delivery process, an interface *DeliveryProcessing* is being designed, which is further implemented by the classes that provide the implementation for specific delivery processes. The interface *DeliveryProcessing* specifies the methods to be implemented by any delivery class, and those are *packageOrder()*, *notifyCustomer()*, *bookDeliveryPartner()* and *deliver()*. The online gift store also has digital gift items like gift coupons, software games, etc., and for delivering these items, most of the methods in the interface *DeliveryProcessing* are not relevant. However, since the interface *DeliveryProcessing* needs to be implemented by any delivery class, delivery classes for digital gift items also need to implement the methods from *DeliveryProcessing* which are not at all relevant to those items.

Identify the SOLID principle that is missing from the design process.

Options :

6406533488878. ✖ Single responsibility principle

6406533488879. ✖ Open-close principle

6406533488880. ✖ Liskov substitution principle

6406533488881. ✔ Interface segregation principle

Question Number : 14 Question Id : 6406531031614 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

An organization provides its employees with a mobile app that allows them to apply for leave. To select dates, it pops out a calendar; however, the calendar cannot highlight the holidays separately. As a result, there is a huge chance that an employee will apply for leaves on holidays. Which of the following heuristics of UI evaluation is missing in the mobile app?

Options :

6406533488882. ✓ Prevent errors

6406533488883. ✗ Recognition over recall

6406533488884. ✗ Provide help

6406533488885. ✗ Flexibility

Question Number : 15 Question Id : 6406531031615 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Suppose a job portal has a class *Applicant* for representing the candidates applying for jobs. Class *Applicant* has a few mandatory fields (like name, contact number, email) and a large number of optional fields (like experience, passport number, PAN number), which necessitates the designer to write a large number of constructors to instantiate the object of *Applicant* class (one constructor with only initializing all the mandatory fields and a number of constructors for different numbers and different combinations of optional fields along with all mandatory fields). Which of the following design patterns can be best suited for solving the problem of creating a large number of constructors?

Options :

6406533488886. ✗ Factory design pattern

6406533488887. ✓ Builder design pattern

6406533488888. ✗ Adapter design pattern

6406533488889. ✗ Facade design pattern

Sub-Section Number :

4

Sub-Section Id :

640653157504

Question Shuffling Allowed :

Yes

Question Number : 16 Question Id : 6406531031618 Question Type : MCQ

Correct Marks : 6

Question Label : Multiple Choice Question

A software company has to develop an automated gardening system to control soil moisture and other chemical components in soil. The developed software should be strongly coupled with the hardware, like sensors, controllers, motors, etc.

The values for the constants in the COCOMO model for the various types of projects are as follows:

- Organic project: $a = 2.4$, $b = 1.05$

- Semi-detached project: $a = 3.0$, $b = 1.12$

- Embedded project: $a = 3.6$, $b = 1.20$

Given the source code of 1,00,000 lines and the developer's monthly salary of Rs. 50,000/-, what is the initial estimated cost for the project?

Options :

6406533488898. ✖ Rs. 2,61,50,000/-

6406533488899. ✖ Rs. 1,51,50,000/-

6406533488900. ✖ Rs. 5,67,00,000/-

6406533488901. ✔ Rs. 4,52,50,000/-

Sub-Section Number :

5

Sub-Section Id :

640653157505

Question Shuffling Allowed :

Yes

Question Number : 17 Question Id : 6406531031616 Question Type : MSQ

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is/are the main advantage(s) of prototype model for software development?

Options :

6406533488890. ✔ It is very effective when the exact form of solution and technical issues are unclear about the software to be developed.

6406533488891. ✔ It is useful to get feedback from customers.

6406533488892. ✖ It reduces the overall development costs.

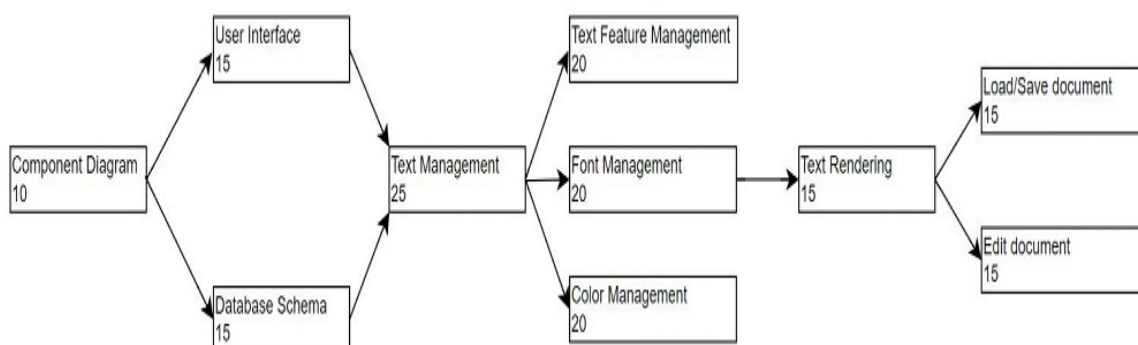
6406533488893. ✖ It ensures that no bugs will appear later in the development cycle.

Question Number : 18 Question Id : 6406531031617 Question Type : MSQ

Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the *activity network diagram* for developing a food delivery mobile app as given below.



Select the correct observations for the given *activity network*.

Options :

6406533488894. ✔ Implementation of *Text Management* starts after the implementations of both *User Interface* and *Database Schema* are completed.

6406533488895. ✖ Implementation of *Text Rendering* starts after the implementation of both *Load/Save document* and *Edit document*.

6406533488896. ✔ The tasks *Text Feature Management*, *Font Management* and *Color Management* can be implemented simultaneously.

6406533488897. ✖ The tasks *Font Management* and *Text Rendering* can be implemented simultaneously.

Sub-Section Number :

6

Sub-Section Id :

640653157506

Question Shuffling Allowed :

No

Question Id : 6406531031598 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (19 to 21)

Question Label : Comprehension

Consider a software organization developing an online gift shop where customers can place an order to send gifts to their friends and family members staying in the same city. The customers can select from a range of gifts and place an order. After placing the order, the store manager contacts the courier service provider via phone call to collect the item from the physical store and schedule delivery.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 19 Question Id : 6406531031599 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Identify the most appropriate user type for the *courier service provider*.

Options :

6406533488820. ✖ Primary users

6406533488821. ✖ Secondary users

6406533488822. ✔ Tertiary users

6406533488823. ✖ Not a valid user

Question Number : 20 Question Id : 6406531031600 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

The developers of the online gift store want to explore the types of gift items to be kept in the catalog. Therefore, they want to gather responses for some specific questions from a large group of people. Which requirement gathering technique would be the most effective for this purpose?

Options :

6406533488824. ✔ Questionnaires

6406533488825. ✖ Interviews

6406533488826. ✖ Focus groups and workshops

6406533488827. ✖ Documentation

Question Number : 21 Question Id : 6406531031601 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider a user story for the online gift shop that describes the following feature:

Feature 1: Search catalog and place order

However, it has been found that implementation of *Feature 1* is not possible within one agile iteration. Therefore the *Feature 1* is subdivided as follows:

Feature 1a: Search catalog and Feature 1b: Place order

Which of the SMART characteristics is fulfilled by the refinement?

Options :

6406533488828. ✖ Specific

6406533488829. ✖ Measurable

6406533488830. ✔ Achievable

6406533488831. ✖ Relevant

6406533488832. ✖ Time-bound

Sub-Section Number :

7

Sub-Section Id :

640653157507

Question Shuffling Allowed :

No

Question Id : 6406531031619 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (22 to 23)

Question Label : Comprehension

A software company has plans to develop a web application and a mobile application that can assess equities and recommend their purchase or sale. Nevertheless, the product might not be competitive in the market due to the abundance of similar products that are already available. This is a potential risk associated with its development.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 22 Question Id : 6406531031620 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question